

Building Agentic AI

Course Summary

Description

This instructor-led program guides developers, data scientists, team leads, and project managers through designing, deploying, and governing AI agents that autonomously plan, reason, reflect, and execute tasks. The course combines 50% lecture and 50% hands-on labs in a cloud-based, zero-install environment.

Objectives

After taking this course, students will be able to:

- Understand design principles of Agentic AI systems
- Build, evaluate, and govern autonomous AI agents
- Integrate Agentic frameworks (LangGraph, LlamaIndex, Bedrock, DSPy, Claude)
- Implement multi-agent collaboration and reflection mechanisms
- Optimize deployments for latency, cost, and evaluation metrics

Topics

- Foundations of Agentic AI
- Reflection, Tools, and Evaluation
- Frameworks, Deployment & Governance
- Final Capstone Project

Audience

This course is designed for Developers, Data Scientists, Team Leads, and Project Managers.

Prerequisite

- Basic understanding of machine learning concepts
- Familiarity with Python (preferred)
- Knowledge of API calls and JSON

Duration

Five days

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Course Outline

I. Foundations of Agentic AI

- A. What is Agentic AI?
- B. From prompt-based AI to autonomous agents
- C. OpenAI's Agentic guidelines and best practices
- D. When and why to use Agentic AI
- E. Design principles: autonomy, reasoning, reflection, planning, safety
- F. Customer Support/Research Assistant Agentic system
- G. Lab 1: Lab environment setup and first "agent loop" using OpenAI & ReAct pattern
- H. Agentic AI workflow: Input → Plan → Act → Reflect → Output
- I. Single vs. multi-step reasoning and tool/memory usage
- J. Lab 2: Build a Research Assistant agent with multi-step reasoning and logging

II. Reflection, Tools, and Evaluation

- A. Reflection and self-evaluation patterns
- B. Improving factuality/style; generating charts/reports reflectively
- C. Lab 3: Build and compare direct vs. reflective summarization agent workflows
- D. Tools in Agentic AI: function calling, schema validation, structured responses
- E. Example tools: email, CRM query, code execution
- F. Lab 4: Convert Python functions to agentic tools
- G. Lab 5: Create and test an "Email Assistant Workflow" (draft, summarize, send emails)
- H. Explore MCP protocol/code execution inside agents

III. Advanced Agentic Patterns

- A. Evaluation metrics, error analysis, debugging (tool errors, plan drift, hallucination)
- B. Workflow optimization and next-step prioritization
- C. Lab 6: Evaluate and benchmark agentic workflows
- D. Multi-agent conversation flows and sequential chatbots
- E. Planning-driven agents for onboarding/blogpost generation
- F. Lab 7: Build an onboarding assistant agent (meeting scheduler, email generator, welcome messaging)

IV. Frameworks, Deployment & Governance

- A. LangGraph & Tavily Integration
- B. LangGraph architecture (nodes/edges/state); agentic search via Tavily
- C. Persistence, streaming, human-in-the-loop feedback
- D. Labs 8: Build an Essay Writer agent with retrieval+reflection on LangGraph
 - 1. Low-Code Agentic Design
- E. Lab 9: Creating business automation agents: ChatGPT + Zapier, Custom GPTs
 - 1. Memory, chaining, and reflection in Zapier
- F. Lab 10: Build a no-code AI CRM agent with Zapier + Custom GPT
- G. Agentic on AWS Bedrock
 - 1. Using Bedrock APIs, CRM/database connection, latency/scalability patterns
 - 2. Guardrails, transactions, compliance
- H. Lab 11: Build a CRM assistant via Bedrock Agent Runtime & Titan Model
- I. Agentic with DSPy and Databricks
 - 1. Declarative agent design, self-improving loops, Databricks enterprise integration
- J. Lab 12: Create a self-improving agent with DSPy, integrate Delta Table analytics
- K. Agentic with Claude
 - 1. Multi-tool coding assistants using Anthropic's Claude; memory/context management
- L. Lab 13: Build Claude-based coding assistant with persistent memory
- M. Voice Agents with Google's ADK
 - 1. Audio Developer Kit for speech input/output, context continuity
- N. Lab 14: Build a live Voice Support Agent (ADK + OpenAI)
- O. Governance with Databricks
 - 1. Tracking agent actions, data lineage, audit logs, HR analytics
- P. Lab 15: Build an HR Governance Agent that audits agent activity logs

V. Final Capstone Project

- A. Enterprise AI Agent Challenge—teams design and demo a multi-agent system (domain: Finance, HR, Real Estate, or Customer Support), showcasing agent reasoning, tool-use, reflection, evaluation, and governance.